I Introduction

Observability tells us if we can figure out the state using measurements of the output.

↳ What sensors are sufficient to do this?

II Definition

Given an LTV system

$$\dot{x} = A(t)x + B(t)u \qquad y = C(t)x + D(t)u$$

A) Unobservable

A state x_0 is unobservable on $[t_0, t_1]$ with $x(t_0) = x_0$ if

$$y(t) = 0 \quad \forall \; t \in [t_0, t_1]$$

B) Unobservable space

The set of unobservable states on $[t_0, t_1]$ is a subspace of $\mathbb{R}^n$

C) Completely Observable

The system is completely observable (C.O.) on $[t_0, t_1]$ if 0 is the the only unobservable state

III Observability Grammian

Consider the system

$$\dot{x}(t) = A(t)x \qquad x(t) = \Phi(t, t_0)x_0$$

$$y(t) = C(t)x \qquad y(t) = C(t)\Phi(t, t_0)x_0$$

We want to know if we can solve

$$y(t) = C(t)\Phi(t, t_0)x_0$$

for unknown x_0

A) Set of Unobservable states

$$\{ x_0 \mid y(t) = \underbrace{C(t)\Phi(t, t_0)}_{\text{Matrix}} \underbrace{x_0}_{\text{vector}} = 0 \; \forall t \in [t_0, t_1] \}$$

$$\Rightarrow x_0 \in N(C(t)\Phi(t, t_0)x_0)$$

B) Observability Grammian

$$W_o(t_0,t_1) = \int_{t_0}^{t_1} \Phi^T(\tau,t_0)\, C^T(\tau)\, C(\tau)\, \Phi(\tau,t_0)\, d\tau$$

↳ ~~If~~ we swap C's for B's it looks like W_R / W_C

C) Theorem

x_0 is unobservable on $[t_0,t_1] \iff x_0 \in N(W_o(t_0,t_1))$

IV LTI Example

Consider the discrete system $x_{k+1} = Ax_k \quad y_k = Cx_k$

$y_0 = Cx_0 \quad \Rightarrow x_1 = Ax_0$

$y_1 = Cx_1 = CAx_0$

$y_2 = Cx_2 = CA^2x_0$

$\vdots$

$y_n = Cx_n = CA^n x_0$

So x_0 is unobservable if

$Cx_0 = 0$

$Cx_1 = 0$

$\vdots$

$Cx_n = 0$

Recalling Cayley-Hamilton Theorem

$$CA^k = \sum_{j=0}^{n-1} \alpha_j A^j \text{ for } k \geq 0$$

Only need to check $x_0 \in N(\Omega)$

where $\Omega = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$

A) Theorem: Observability Matrix Nullspace

$N(W_o(t_0,t_1)) = N(\Omega) \;\forall\; t_1 \geq t_0$

B) Complete Observability

Given $\dot{x} = Ax$, $y = Cx$, the system is completely observable iff $\text{rank}(\Omega) = n$ (iff $N(\Omega) = \{0\}$).

Note the symilarity:

$$\Omega^T = C^T \; A^T C^T \; A^{T^2} C^T \; \dots \; (A^{n-1})^T C^T \;]$$

where $\text{rank}(\Omega) = \text{rank}(\Omega^T)$

Assume alternate definition

$$\dot{x} = \underbrace{A^T}_{\bar{A}} x + \underbrace{C^T}_{\bar{B}} u$$

(A, C) is completely observable iff $(\bar{A}, \bar{B})$ is completely controllable where

$$\bar{A} = A^T \quad \bar{B} = C^T$$

A) Example

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \quad C = [1 \;\; -1]$$

$$n = 2 \quad \Omega = \begin{bmatrix} C \\ CA \end{bmatrix} \quad CA = [-1 \;\; 1]$$

$$\Omega = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \Rightarrow \text{rank}(\Omega) = 1$$

The system is not completely observable

What is $N(\Omega)$

$$\Omega x = 0 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} x_1 = x_2 \\ x_2 = x_1 \end{matrix}$$

$\Rightarrow N(\Omega) = \text{span}\left\{\begin{bmatrix}1\\1\end{bmatrix}\right\}$

VI General A-Invariant Property

For our case: If we start in null space, do we stay there?

A) Property: The null space of the observability matrix is A-invariant

Let $x_0 \in N(\Omega) \Rightarrow \Omega x_0 = 0 = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} x_0 = 0$

Take a look at $\Omega A x_0$

$$\Omega A x_0 = \begin{bmatrix} CAx_0 \\ CA^2 x_0 \\ \vdots \\ CA^n x_0 \end{bmatrix} = 0 \text{ as defined}$$

Using Caley Hamilton

$$CA^n x_0 = \sum_{i=0}^{n-1} \alpha_i CA^i x_0$$

This means that $x(t) \in N(\Omega)\ \forall t > t_0$

$\Rightarrow$ If you start in the null space you stay in the null space (where there is no control).

VII Observable Decomposition

We can produce an observable decomposition

Given $\dot{x} = Ax$, $y = Cx$

let $v_1, \dots, v_d$ form a basis for $N(\Omega)$

let $u_1, \dots, u_{n-d}$ be vectors that compliment $v_1 \dots v_d$ to form a basis for $\mathbb{R}^n$

Let $T = [u_1, \ldots, u_{n-d}, v_1, \ldots, v_d]$ Reverse of controllable decomposition

$\hat{A} = T^{-1} A T \quad \hat{C} = CT$

$$\hat{A} = \begin{bmatrix} \hat{A}_{11} & 0 \\ \hat{A}_{21} & \hat{A}_{22} \end{bmatrix} \quad \hat{C} = \begin{bmatrix} \hat{C}_1 & 0 \end{bmatrix}$$

$\hat{A}_{11}$: $n-x \times n-x$ $\qquad$ $\hat{A}_{21}$: $d \times n-d$

$\hat{A}_{22}$: $d \times d$ $\qquad$ $\hat{C}_1$: $p \times n-d$

$$\hat{x} = \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix}$$ The pair $(\hat{A}_{11}, \hat{C}_1)$ is C.O.

$x_1 \rightarrow$ observable

$x_2 \rightarrow$ Unobservable